Hydrocarbon - 1

Carbon Free Radical :

$\dot{C}H_3$ → Paramagnetic specie

C : (1) (2) (3) (4), Free Radical
- → Highly unstable
- → sp^2 hybridised
- → Incomplete octet
- → Trigonal Planer

→ Reactions shown by Carbon free Radical
- → Combination
- → Disproportionation
- → Rearrangement (Not in JEE Syllabus)

(Hence, for JEE, C° would never Rearrange)

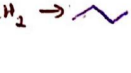
(I) Combination Reaction :→

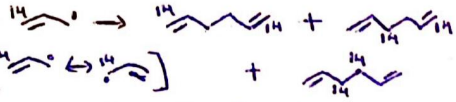
→ 1° or 2° free radicals can easily combine with each other.

1) $2\,\dot{C}H_3 \rightarrow CH_3-CH_3$

2) $2\,CH_3-\dot{C}H_2 \rightarrow$

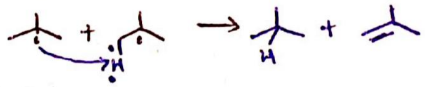
3)

4) 

5) 

(II) Disproportionation Reaction :→

→ 3° free radicals can't combine due to huge steric repulsion, thus they disproportionate.

Eg: 

Wurtz Reaction :

$$2\,R-X \xrightarrow[\text{in dry } Et_2O]{Na} R-R$$

Mechanism ⇒

$$R-X \xrightarrow[(1^\circ,2^\circ)]{Na} (R^\circ) + Na^+X^- \xrightarrow{(3^\circ)} (\text{Alkane + Alkene})$$

$(R^\circ) \rightarrow (R-R)$

Important Points ⇒

(i) Free Radical intermediate

(ii) R·O·R : R-I > R-Br > R-Cl > R-F

(iii) Dry ether taken bcz in case of moist ether, entire Na would react with H_2O.

(iv) Symmetrical even carbon alkanes obtained as major product.

(v) CH_4(methane) can never be obtained.

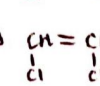
Wurtz ($R-X + Na \rightarrow R-R$) (Only Alkyl)
- Fitting ($Ph-X + Na \xrightarrow{\Delta} Ph-Ph$) (Only Aryl)
  - Ullmann (Cu/Δ): $PhX + Cu \rightarrow Ph-Ph$
  - Frankland (Zn/Δ): $R-X \xrightarrow[\Delta]{Zn} R-R$; $Ph-X \xrightarrow[\Delta]{Zn} Ph-Ph$

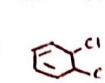
Wurtz-Fitting Rxn

$$R-X + PhX \xrightarrow[\Delta]{Na} R-Ph \text{ (major)} + RR + Ph-Ph \text{ (Minor)}$$

1) 

2) 

3) 

4) 

5) 

(Dimerisation took place bcz [11] is anti Aromatic)

6) 

7) 

Corey House Synthesis :

→ This Rxn is used to synthesis unsymmetrical alkanes that we were unable to obtain by Wurtz Rxn.

$$R-X \xrightarrow[\text{(ii) } Cu-X]{\text{(i) } Li} R-R'$$
(iii) $R'-X$

$$CH_3-CH_2-Br \xrightarrow[\text{(ii) } CuBr]{\text{(i) } Li} CH_3-CH_2-CH_3$$
(iii) CH_3Cl

Mechanism
(i) $RX + 2Li \rightarrow R^\ominus Li^\oplus$
(ii) $2RLi + CuX \rightarrow R_2CuLi$ (called Gilliman's Reagent)
(iii) $R_2CuLi + R'X \rightarrow R-R'$

Important Points ⇒
- → Carbanion Intermediate
- → Gilliman Reagent obtained during Rxn
- → R·O·R for R' : 1° R' > 2° R' > 3° R'

Kolbe's Electrolysis :

→ Electrolysis of Carboxylate salts is done.

Anode: Electrode on which Anion goes

Cathode: Electrode on which Cation goes

Anode:

(i) $R-\overset{O}{\overset{\|}{C}}-O^\ominus \xrightarrow{-e^-} R-\overset{O}{\overset{\|}{C}}-O^\circ \rightarrow R^\circ + CO_2$

$R^\circ \rightarrow (R-R)$

Cathode:

(ii) $H-OH \xrightarrow{e^-} \frac{1}{2}H_2\uparrow + NaOH$

Important Points ⇒
- → Free Radical Mechanism
- → CO_2 liberated at Anode and H_2 liberated at cathode
- → CH_4 can not be synthesised by this process
- → Alkane formed can also be gas.
- → pH of soln ↑ses (due to NaOH) during rxn.

1) $CH_3COONa \xrightarrow{\text{Electrolysis}} CH_3-CH_3$

2) $CH_2(CH_2-COONa)_2 \rightarrow CH_2(CH_2)_2$

3) $CH=COONa \rightarrow CH \equiv CH$

4) $HCOONa \rightarrow H_2\uparrow + CO_2\uparrow$

5) $CH_3-SO_3Na \rightarrow CH_3-CH_3 + SO_3\uparrow$

Decarboxylation :

$$R-\overset{O}{\overset{\|}{C}}-ONa \xrightarrow[CaO]{NaOH} RH + Na_2CO_3$$

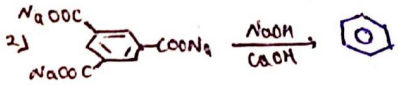
→ Also called Sodalime decarboxylation / Oakwood Degradation Rxn.

→ Carbanion Intermediate Imp

↳ Rate of Rxn ∝ Stability of Carbanion

→ NaOH : CaO ⇒ 3 : 1

1) $CH_3-COONa \xrightarrow[CaO]{NaOH} CH_4$

2) 

3) Compare rate of decarboxylation for :

$CH_3^\ominus$ (sp^3) < $CH_2=CH^\ominus$ (sp^2) < $HC\equiv C^\ominus$ (sp)

[Jitna zyada % character, utni zyada electronegativity, utna stable −ve charge]

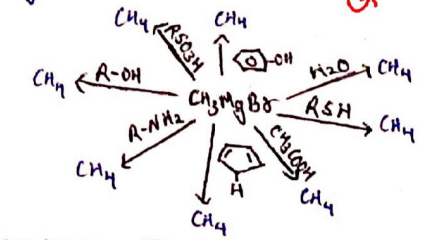
Reaction with Grignard Reagent :

Acid Base Rxn of Grignard

$R^{\delta-}MgX + E_n^{\delta+}-H \rightarrow R-H$

Substitution Rxn of Grignard

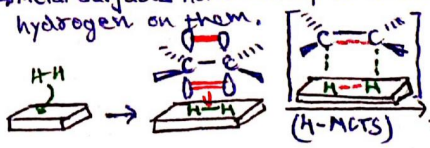
$R-X + {}^{\delta-}RMgX \rightarrow R-R'$



[Examples of Acid-Base Rxn of G.R]

Hydrogenation :

→ Metal surfaces have ability to adsorb gaseous hydrogen on them.



→ Hydrogenation is a surface phenomena.

$$CH_2=CH_2 \xrightarrow[\text{or Ni or Ra-Ni}]{H_2/Pt \text{ or } Pd} CH_3-CH_3$$
(Ranney Nickel)

→ Rate of Hydrogenation ∝ $\frac{1}{\text{Steric Crowding}}$

→ It is Syn-addition Reaction.

Stereochemistry :→

Jo first alkene, wapsa addition ⇒ Bada wala naam (Erythrose)

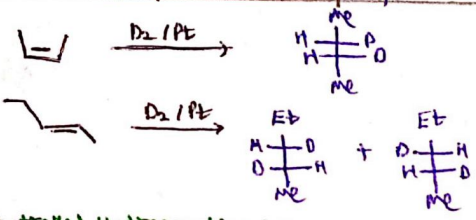


Jo first alkene uska opp. addition ⇒ Chota wala naam (Threose)



(Valid for Entire Organic Chemistry)

Type of Alkene	Type of Addition	Type of Product
Cis (+)	Syn (+)	Erythrose (+)
Cis (+)	Anti (−)	Threose (−)
Trans (−)	Syn (+)	Threose (−)
Trans (−)	Anti (−)	Erythrose (+)



Controlled Hydrogenation :→

→ Use of $BaSO_4/CaCO_3$ like poisons with catalysts stops red^n after 1 step.

$$HC\equiv CH \xrightarrow[Pd-BaSO_4]{H_2} CH_2=CH_2$$

Lindlar's catalyst

Reactions of Alkane :

1] Substitution → Halogenation, Nitration, Sulphonation.
2] Isomerisation
3] Aromatisation
4] Combustion / Controlled Oxidation
5] Pyrolysis.

(I) Halogenation :→

$$R-H + X_2 \xrightarrow{h\nu} R-X + H-X$$

Mechanism ⇒

(i) $X_2 \xrightarrow{h\nu} 2X^\cdot$ (Initiation)
(ii) $R-H + X^\cdot \xrightarrow{(rds)} R^\cdot + HX$ (slowest step) } (Propogation)
(iii) $R^\cdot + X_2 \rightarrow RX + X^\cdot$ }
(iv) $R^\cdot + R^\cdot \rightarrow R-R$
$R^\cdot + X^\cdot \rightarrow R-X$
$X^\cdot + X^\cdot \rightarrow X-X$ } (Termination)

Fluorination (F$^\cdot$)
→ Highly unstable F$^\cdot$
→ Highly exothermic / explosive rx^n

$$CH_4 \xrightarrow[h\nu]{F_2} C_{Black} + HF$$

Iodination (I$^\cdot$)
→ Highly stable I$^\cdot$
→ Least reactive / highly reversible rx^n

$$R-H + I_2 \xrightarrow[h\nu]{\Delta} R-I + HI$$

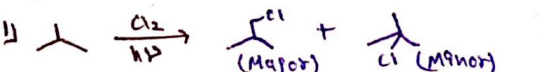
Oxidising agents like HIO_3 & HNO_3 added so that they convert HI into I_2 & eq^m shifts in forward direction.

Chlorination (Cl$^\cdot$)
→ Less stable
→ More Reactive
→ Less selective
→ Rx^n Ratio : 1 : 3.8 : 4.5 (1°) (2°) (3°)

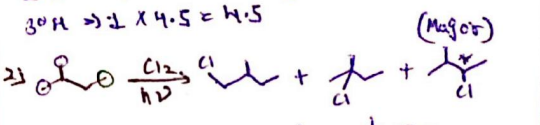
Bromination (Br$^\cdot$)
→ More stable
→ Less Reactive
→ More selective
→ Rx^n Ratio : 1 : 81 : 1600 (1°) (2°) (3°)

→ Thus in case of Br$^\cdot$, major product will always be decided by degree of free radical

→ But, in case of Cl$^\cdot$, major product often decided by Rx^n Ratio data.

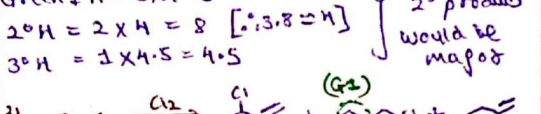
1] 

1° H ⇒ 9 × 1 = 9
2° H ⇒ 0 × 3.8 = 0 [∵ 9 > 4.5]
3° H ⇒ 1 × 4.5 = 4.5

2] 

Blue 1° H = 6 × 1 = 6
Green 1° H = 3 × 1 = 3
2° H = 2 × 4 = 8 [∵ 3.8 ≈ 4]
3° H = 1 × 4.5 = 4.5

} 2° product would be major

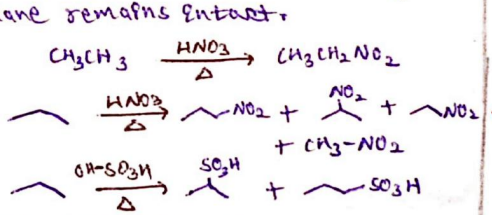
3] 

Total products = 2 + 2 + 1 = 5
(O.I) (G.I)

(II) Nitration & Sulphonation :→

→ HNO_3 is strong reagent, thus alkane breaks while rx^n occurs, but in case of H_2SO_4 alkane remains intact.

$$CH_3CH_3 \xrightarrow[\Delta]{HNO_3} CH_3CH_2NO_2$$

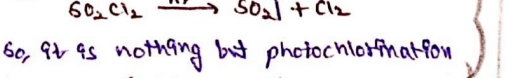


Reed's Reaction :

Here, alkane is reacted with SO_2Cl_2 & $h\nu$.

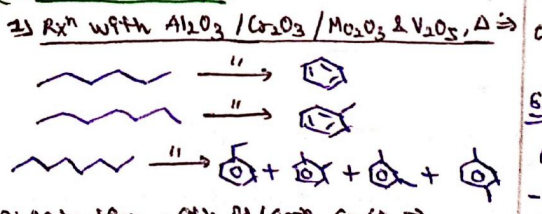
$$SO_2Cl_2 \xrightarrow{h\nu} SO_2\uparrow + Cl_2$$

So, it is nothing but photochlorination

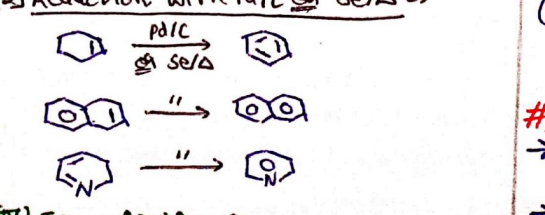


(III) Aromatisation :→

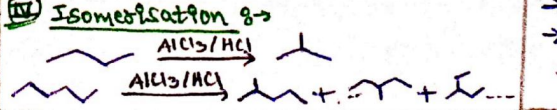
1] Rx^n with $Al_2O_3/Cr_2O_3/Mo_2O_3$ & V_2O_5, Δ ⇒



2] Reduction with Pd/C & Se/Δ ⇒



(IV) Isomerisation :→



(V) Combustion :→

Complete Combustion

$$C_xH_y + \left(x + \frac{y}{4}\right)O_2 \rightarrow x\cdot CO_2 + \frac{y}{2}H_2O$$

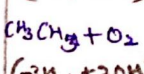
Incomplete Combustion

$$CH_4 + O_2 \rightarrow C(s) + 2H_2O$$

(VI) Controlled Oxidation :→

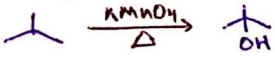
$CH_4 + O_2 \xrightarrow[523\,K]{Cu} CH_3-OH$ (−1H, +1OH)

$CH_4 + O_2 \xrightarrow{Mo_2O_3} CH_2(OH)_2 \Rightarrow HCHO$ (−2H, +2OH)

$CH_3CH_3 + O_2 \xrightarrow[\Delta]{Mn(CH_3COO)_2}$  (−3H, +3OH)

Rx^n with $KMnO_4$

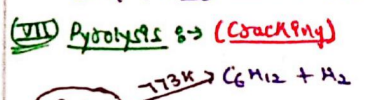
Only $KMnO_4$ can oxidise a 3° alkane



Rx^n with Steam

$$CH_4 + H_2O \xrightarrow{\Delta} CO + 3H_2$$

(VII) Pyrolysis :→ (Cracking)



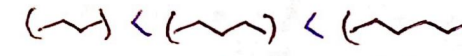
$$C_{12}H_{26} \xrightarrow{773K} C_7H_{16} + C_5H_{10} + \text{other}$$

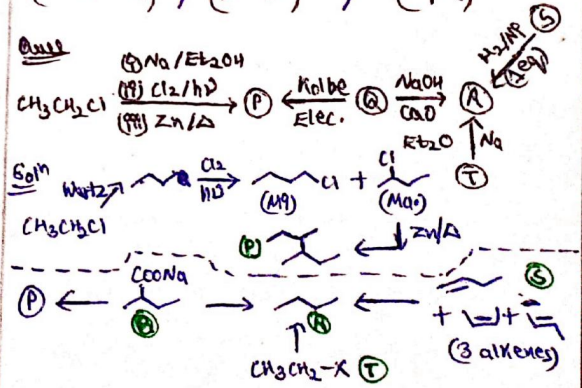
Physical Properties of Alkane :

$C_1 - C_4$ ⇒ Gas
$C_5 - C_{17}$ ⇒ Liquid
C_{18} ⇒ Solid

B.P ∝ Molecular Weight
B.P ∝ 1 / Branching

Eg : Compare Boiling Points of Alkanes :





Carbocation Rearrangement :

→ Carbocations can rearrange to attain more stability.
→ Takes place in 1,2-shifts
→ Rearrangement would occur only if cation formed after it is more stable than previous one.

Hydrocarbon - 2

→ The stability can be due to
- → Lone pair atom on adjacent position
- → Resonance
- → Hyperconjugation (α-H)
- → ↓ in Angle strain (Ring Expansion)

→ If lone pair atom is present and back bonding is possible, then only following migratory order is followed:

[H > Ar > 3° > 2° > 1° > -CH₃]

1) (3 α-H) ⇌ (1 α-H)

2) No Rearrangement

3) No Rearrangement (Ring Expansion)

4) [Back Bonding > Hyperconjugation]

5)

6) Exceptionally stable due to π-Resonance. No Rearrangement.

7) No Ring expansion bcz (6) stability > (5) stability.

Ring will unusually contract or expand only when carbocation is being given b/cse of back bonding stability for it.

8) (Ring Contraction)

9) (6 membered) ⇌ (7 membered) (Unusual Ring Expansion)

[Now we Discuss M.O.P of Alkenes]

Dehydration of Alcohols:

R-OH $\xrightarrow[\text{conc. } H_3PO_4 \text{ or } KHSO_4]{\text{conc. } H_2SO_4 / \Delta}$ Alkene

Mechanism ⇒

R-OH $\xrightarrow{H^+}$ $R-\overset{H}{\underset{H}{O}}-H$ $\xrightarrow[-H_2O]{(S_N1)}$ $R^{\oplus}$ $\xrightarrow{-H^+}$ Stable Alkene

Important Points ⇒

(i) Carbocation Intermediate
(ii) Rearrangement Possible
(iii) Unimolecular Elimination (E1) Mechanism
(iv) [Rate of Rxn ∝ Stability of Carbocation]
(v) Stable alkene would be major product

1) (Minor) + (Major)

2) (Pinacole to Pinacolone) (Major) + + +

3)

Bimolecular Elimination Reactions:

→ Here, Base reacts with acidic α-H of a leaving group substituted alkane.

Base:
- Moderate (Alc. KOH, RO^-): V. Good lg → Moderate lg → Poor lg ($-F, -OH, -OR, -NR_3, -SR_2$) — Most stable alkene is major
- Very Strong ($t\text{-BuO}^-$ > O^-, NH_2^-, etc): (Any lg) — That product is major for which δ- stability of α-position of lg is better.

Eg:
- EtO^-/Δ
- (i) AgF
- (ii) EtO^-/Δ (δ1 < δ2)
- $t\text{-BuO}^-/\Delta$
- (i) AgF
- (ii) O^-/Δ

→ E2 is anti-elimination. Would take place only when α-H & lg are anti to each other.

- EtO^-/Δ (Most stable product)
- EtO^-/Δ Would react only with dash H i.e. (⟵H)
- EtO^-/Δ No Rxn, because no anti α-H present here.

→ Moderate bases cannot eliminate resonance stabilised -lg. For that, strong base required.

$CH_3-CH(Cl)_2 \xrightarrow[\Delta]{\text{Alc. KOH}} CH_2=CH-Cl$ (Partial double bond character due to Reso.)

$CH_3-CH(Cl)_2 \xrightarrow[(ii)\ NaNH_2/\Delta]{(i)\ \text{Alc. KOH}} CH\equiv CH$ (Second elimination due to NH_2^-)

Also, -H of terminal alkynes is very acidic and can also show Acid-Base Rxn.

$HC\equiv CH \xrightarrow{NaNH_2} HC\equiv C^{\ominus}\ Na^{\oplus} + NH_3$
$\downarrow CH_3-I$
$HC\equiv C-CH_3 + NaI$

NOTE: Unimolecular Conjugate Base elimination (E1-CB) would occur when there is EWG group at α-position & poor lg.

Dehalogenation:

- Zn/Δ → (Unstable due to repulsion) → $+I_2$
- NaI, Acetone ($\gamma=0$)

→ For this rxn, if both -Cl are anti, then yield would be high.

→ Therefore, following isomer of Gammexane would show very high yield for benzene.

$\xrightarrow[\text{Acetone}]{NaI}$

Electrophilic Addition Reaction:

(Alkene)
- E^+ without lp → ① Classical carbocation ② Rearrangement ③ Both syn & Anti addition
- Rx with lp → ① Non-Classical Carbocation (NCC) ② No Rearrange. ③ Only Anti Addition

The counter ion of E^+ attacks on NCC from opp. side due to which always Anti-addⁿ occurs.

Eg: E^+ without lp	E^+ with lp
HCl, HBr, H_2O/H^+	Br_2, NOCl (NO^+, Cl^-), $Hg(OAc)_2$

Hydrohalogenation ⇒

1) $\xrightarrow{HBr}$

Hydration of Alkene ⇒

2) $\xrightarrow[H_2O]{H^+}$ $\xrightarrow{-OH}$

3) $\xrightarrow[H_2SO_4]{\text{dil.}}$

4) $\xrightarrow[H_2SO_4]{\text{conc.}}$ SO_3H ↓ (White ppt)

Rxn with conc. H_2SO_4 is used as a test for unsaturation as it gives ppt.

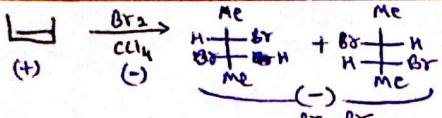
5) $CH_3-C\equiv CH \xrightarrow[(ii)\ HI]{(i)\ HCl} CH_3-\underset{Cl}{\overset{I}{C}}-CH_3$

6) $CH_3-C\equiv CH \xrightarrow[H_2O]{H^+}$ $CH_3-\overset{OH}{C}=CH_2 \rightleftharpoons CH_3-\overset{O}{\overset{\|}{C}}-CH_3$ (Propanone)

7) $CH_3-CH_2-CH=CH_2 \xrightarrow[CCl_4]{Br_2}$ (Brown) → (Colourless)

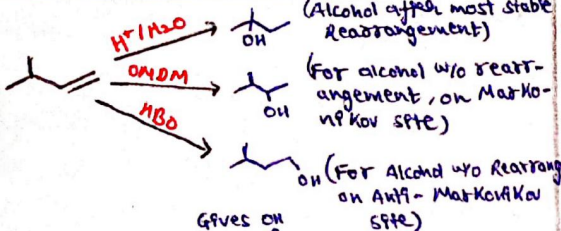
Rxn of Bromine is used as a test for unsaturation as sol^n colour changes from Brown to colourless.

8) $\xrightarrow[CCl_4]{Br_2}$ [Trans + Anti ⇒ Erythro (-) (-) (+)]

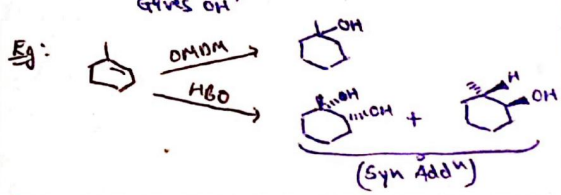
9) 

10) $CH_3-C\equiv CH \xrightarrow[CCl_4]{Br_2} CH_3-\underset{Br}{\overset{Br}{C}}-\underset{Br}{\overset{Br}{CH}}$

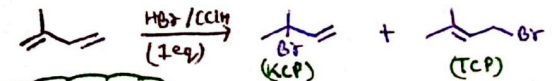
Oxymercuration-Demercuration (OMDM) and Hydroboration Oxidation (HBO) ⇒



Eg:



Kinetically Controlled Product (KCP) & Thermodynamically Controlled Product (TCP) :



If temp. is not mentioned then assume it Room temp. and there TCP would be major.

→ Kinetically fastest product formed.
→ Major at very low temp. (−80°C)

→ Thermodynamically most stable product formed.
→ Major at Room/high temp. (80°C)

Kucherov Reaction :

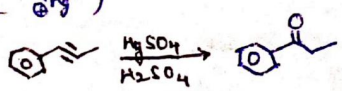
$CH_3-C\equiv CH \xrightarrow{H^+/H_2O} CH_3-\overset{O}{\overset{\|}{C}}-CH_3$

It has low yield bcz during rxn, there (+) comes on doubly bonded carbon which is highly unstable.

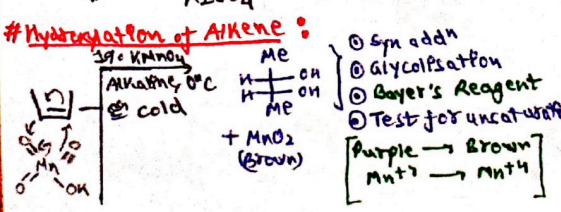
So, to avoid this, we must go through a path where less is formed.

$CH_3-C\equiv CH \xrightarrow[H^+/H_2O]{1\% HgSO_4} CH_3-\overset{O}{\overset{\|}{C}}-CH_3$ (High yield)

$(CH_3-C\equiv CH\overset{\oplus}{H}g)$

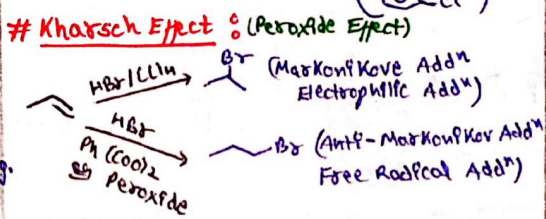
Eg: 

Hydroxylation of Alkene :



Oxymercuration-Demercuration (OMDM) and Hydroboration Oxidation (HBO) ⇒

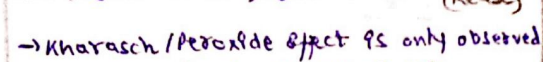
Kharasch Effect : (Peroxide Effect)

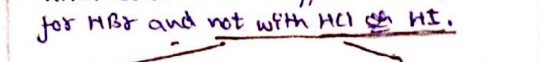


Mechanism ⇒

(i) $Ph-\overset{O}{\overset{\|}{C}}-O-O-\overset{O}{\overset{\|}{C}}-Ph \xrightarrow{h\nu} Ph\dot{C}OO^{\bullet} \rightarrow Ph^{\bullet} + CO_2$

(ii) $Ph^{\bullet} + H\dot{B}r \rightarrow Ph-H + Br^{\bullet}$

(iii)  (Most stable radical)

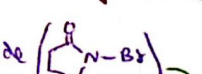
(iv)  (Reuse)

→ Kharasch / Peroxide effect is only observed for HBr and not with HCl & HI.

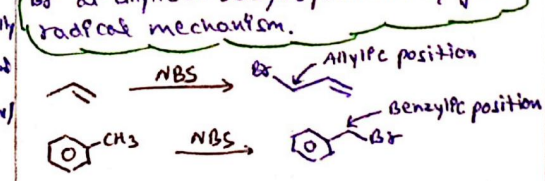
Because Breaking of H-Cl in (iv) step would be highly endothermic & non-spontaneous

Because $I^{\bullet}$ is itself very stable, so rxn would need external energy for attack of $I^{\bullet}$ on

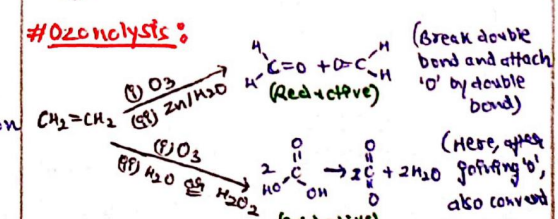
Reaction with NBS :

NBS ⇒ N-Bromo Succinimide 

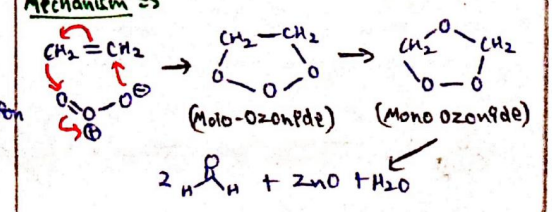
Source of $Br^{\bullet}$ and substitutes $Br^{\bullet}$ at allylic & benzylic positions by free radical mechanism.



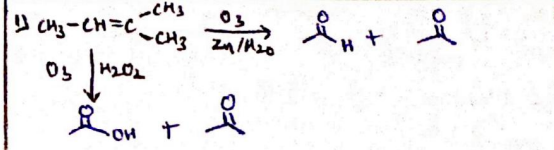
Ozonolysis :



Mechanism ⇒

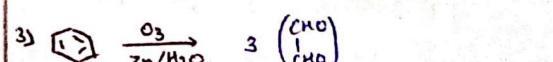


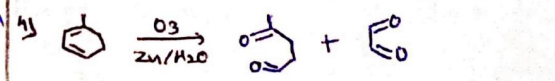
Ozonolysis of Alkene :

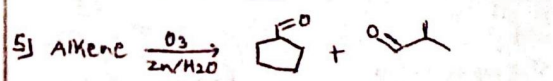


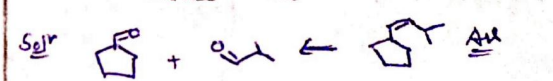
2) Alkene $\xrightarrow[Zn/H_2O]{O_3}$ Propanal + Acetone (Find Alkene)

Sol^n 

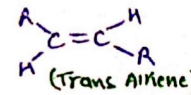
3) 

4) 

5) Alkene $\xrightarrow[Zn/H_2O]{O_3}$ 

Sol^n 

Birch Reduction :

$R-C\equiv C-R \xrightarrow[4n\ Hg,\ NH_3]{Na}$  (Trans Alkene)

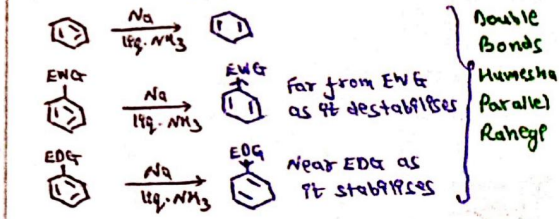
(Blue in colour)

→ Hydrogenation (H_2/Pd) ⇒ Cis Alkene
Birch Red^n (Na/liq NH_3) ⇒ Trans Alkene

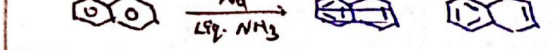
→ Birch Reduction is not shown by terminal alkyne. It rather performs Acid-Base Rxn.

$R-C\equiv C-H \xrightarrow[liq.NH_3]{Na} R-C\equiv C^{\ominus}N\dot{a} + \frac{1}{2}H_2\uparrow$

→ Birch reduction converts aromatic compounds into non-conjugated cyclo-dienes.



→ Birch Reduction only reduced one ring at a time to produce non-conjugation



Hydrolysis of Carbides :

1) $CaC_2 \xrightarrow{2H_2O} Ca(OH)_2 + C_2H_2$ (HC≡CH)

2) $Mg_2C_3 \xrightarrow{4H_2O} 2Mg(OH)_2 + C_3H_4$ ($H_2C=C=CH_2$)

3) $Be_2C \xrightarrow{4H_2O} 2Be(OH)_2 + CH_4$

4) $Al_4C_3 \xrightarrow{12H_2O} 4Al(OH)_3 + 3CH_4$

Reaction with Red Hot Fe Tube :

1) $3\ CH\equiv CH \xrightarrow[Fe\ Tube]{Red\ Hot}$ 

2) $3\ CH_3-C\equiv CH \xrightarrow{\Delta}$ 

3) $3CH_3-C\equiv C-CH_3 \xrightarrow{\Delta}$

4) $3\ Et-C\equiv C-CH_3 \xrightarrow{\Delta}$ 